

4 GOVREPORT Dataset

We introduce a new large-scale dataset, GOVREPORT, containing 19,466 long reports published by U.S. Government Accountability Office (GAO)⁴ to fulfill requests by congressional members, and Congressional Research Service (CRS)⁵, covering researches on a broad range of national policy issues. A human-written summary is provided along with each report. During data collection, we remove boilerplates from crawled files, and keep the section and paragraph structure of the documents and summaries. Additional data cleaning and processing details are in Appendix A.

We obtain 12,228 GAO reports and 7,238 CRS reports of high quality evidenced by human inspection of 200 parsed reports. Collected GAO reports and CRS reports have on average 6.9 and 4.6 sections, respectively. We split train, validation and test set by publication date on each dataset, and end up with 17519 training samples, 974 validation documents, and 973 test samples.

Notably, summaries of GAO reports are written by experts, and are often structured into three aspects in order: “Why GAO did this study”—motivation and problem(s) under discussion, “What GAO found”—findings of the report, and “What GAO recommends”—suggestions and solutions to the problem(s). All but three GAO summaries include “What GAO Found”. The percentages of GAO summaries that contain “Why GAO did this study” and “What GAO recommends” are 94.8% and 29.0%. For comparison, structured summaries are also observed on PUBMED (Cohan et al., 2018) samples. Though they do not contain explicit aspect labels, the summaries can often be broken down into “Introduction”, “Methods”, “Results”, and “Conclusion” via keyword matching. Details about keyword choices for each aspect are provided in Table 11 in Appendix D.

Comparison with Existing Long Document Summarization Datasets. In Table 2, we compare GOVREPORT with several existing long document summarization datasets, including PUBMED and ARXIV (Cohan et al., 2018) that consist of scientific publications; BILLSUM (Kornilova and Eidelman, 2019), a collection of congressional bills; and BIGPATENT (Sharma et al., 2019), a corpus of

Dataset	# Doc	Summary		Doc # word	Comp. # sent	Den.
		# word	# sent			
PUBMED	133,215	202.4	6.8	3049.0	16.2	5.8
ARXIV	215,913	272.7	9.6	6029.9	39.8	3.8
BILLSUM	23,455	207.7	7.2	1813.0	13.6	4.1
BIGPATENT	1,341,362	116.5	3.7	3573.2	36.3	2.4
GOVREPORT	19,466	553.4	17.8	9409.4	19.0	7.3

Table 2: Statistics of GOVREPORT and existing long document summarization datasets. **Comp.:** compression ratio, **Den.:** extractive fragment density (Grusky et al., 2018). All values are mean over the whole dataset except for the “# Doc” column. Documents and summaries in GOVREPORT are significantly longer.

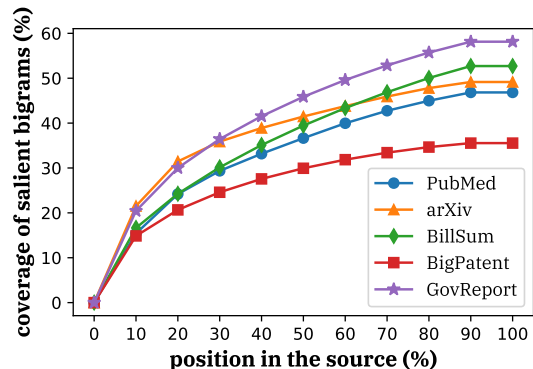


Figure 2: Percentage of unique salient bigrams accumulated from the start to X% of the source. Key information is spread over the documents in GOVREPORT, highlighting the importance of understanding longer text.

U.S. patent documents.

First, *documents and summaries in GovReport are significantly longer than prior datasets*. Next, we inspect the distribution of summary-worthy bigrams in the source by dividing each document into ten equisized partitions. For each partition, we count the occurrence of unique bigrams that also appear in the reference, accumulated from the start of the document to the end of the partition. Fig. 2 shows that *key information is spread throughout documents in GOVREPORT*, with new salient bigrams being steadily added as more content is consumed. For ARXIV and BIGPATENT, only about 10% of new salient bigrams are accumulated in the second half of the documents, reflecting the heavy positional bias in these two datasets. In contrast, in GovReport and BILLSUM, more than 18% of new summary-worthy bigrams appear in the later half of the articles, showing a more even distribution. A similar trend is observed on unigrams. However, BILLSUM has the shortest documents among the five datasets.

⁴www.gao.gov

⁵crsreports.congress.gov

5 Summary Evaluation with Cloze QA

This work aims to evaluate whether processing more text improves both informativeness and faithfulness of abstractive summaries. In addition to ROUGE (Lin, 2004) and human evaluation, we extend existing QA-based metric (Eyal et al., 2019) and consider an entailment-based scorer.

QA-based Evaluation. We present a new faithfulness evaluation metric by extending the APES score (Eyal et al., 2019). We follow APES to construct a set of **cloze questions**, $\{q\}$, from each reference summary by masking entities. Events, dates, and numbers are also masked, as they are prevalent in our data. Each masked phrase becomes the gold-standard answer a_{ref} for a question q . We do not generate natural language questions (Durmus et al., 2020; Wang et al., 2020a), due to the lack of accurate question generation models for the domains of government reports and scientific papers.

QA models are trained by reading a question and a **context** to label the answer span in the context. We construct context by greedily selecting sentences that maximize the improvement of ROUGE-2 recall when compared with the reference summary. If the answer a_{ref} cannot be found in the context, the sample is excluded from training. We train all QA models by fine-tuning BERT (Devlin et al., 2019) to predict the answer span.

To evaluate the faithfulness of a system summary, **APES** uses the QA model to read the summary and a question q to label an answer a_{sys} . It calculates a unigram F1 score by *comparing* a_{sys} and a_{ref} . Different from APES, we further use the QA model to read the context (sentences selected from the source) and give an answer a_{cxt} to the question q . We compute a unigram F1 by *comparing* a_{sys} and a_{cxt} , denoted as **APES_{src}**. Given that existing summarization models rarely rewrite names or numbers correctly, our metric can better capture faithfulness by using a gold-standard answer constructed from the source article than from the human-written abstract.

To **extract entities and events**, we deploy a state-of-the-art IE framework, OneIE (Lin et al., 2020) on GOVREPORT. On PubMed, we re-train OneIE on Genia 2011 (BioNLP, 2011) and 2013 (BioNLP, 2013), and PubMed (Wei et al., 2019) datasets to extract domain-specific entities and events, such as entities of *Gene* and *Disease*. We additionally include numbers and dates extracted by spaCy (Honnibal and Montani, 2017).

Entailment-based Evaluation. We further consider **FactCC** (Kryscinski et al., 2020), which evaluates factual consistency of a system summary by predicting an entailment score between the source and the summary. We reproduce their method on our datasets.

Additional details for implementing the evaluation models and the entity extraction models are given in Appendix B.

6 Experimental Results

In this section, we start with describing training details in § 6.1. We then compare attention variants on documents of the same length (§ 6.2) and study whether reading more text can generate more informative summaries (§ 6.3). We further report human evaluation on summary informativeness and faithfulness as well as automatic faithfulness scores (§ 6.4). Finally, we investigate whether automatic metrics correlate with human judgment (§ 6.5).

6.1 Training Details

We fine-tune BART (Lewis et al., 2020) for all experiments. We implement our models with PyTorch (Paszke et al., 2019) and Fairseq (Ott et al., 2019). Additional position embeddings are initialized randomly for models that handle longer inputs. The learning rate is set to 1×10^{-4} and learning rate warm-up is applied for the first 10,000 steps. Adafactor (Shazeer and Stern, 2018) optimizer with a gradient clipping of 0.1 is used. All models are trained on two Quadro RTX 6000 GPUs with 24GB memory or one Quadro RTX 8000 with 48GB memory. We set a batch size of 2 per step and accumulate gradient every 32 steps. During test, we adopt a beam size of 4 and a length penalty of 2 (Wu et al., 2016) on all datasets.

6.2 Comparing Attention Variants

Comparisons. We first experiment with articles that are all truncated at 1024 tokens. For encoder attentions, we consider the following variants: (1) sliding WINDOW; (2) adaptive span (ADASPAN); (3) GLOBAL tokens; (4) STRIDE; (5) RANDOM tokens; (6) Linformer (LIN.); (7) locality sensitive hashing (LSH); and (8) SINKHORN. We ensure models are comparable by setting hyperparameters to satisfy $w = \hat{w} = k = lb_l = 2b_s = 256$, so that models have similar memory complexity. For LSH attentions, we select $l = 4$ rounds of hashing. Following prior work (Zaheer et al.,

System	GovReport (new)			PubMed		
	R-1	R-2	R-L	R-1	R-2	R-L
FULL	52.83	20.50	50.14	45.36	18.74	40.26
Encoder variants w/ full enc-dec attn.						
<i>I. Fixed Patterns</i>						
WINDOW	50.78	18.59	48.10	42.74	16.83	37.96
+ GLOBAL	51.24	19.01	48.58	43.44	17.07	38.55
+ STRIDE	51.53	19.14	48.68	43.73	17.25	38.82
+ RANDOM	51.49	18.90	48.75	43.38	16.87	38.45
ADASPAN	50.76	18.69	48.13	43.42	17.16	38.60
+ GLOBAL	50.33	18.56	47.80	43.24	17.01	38.42
+ STRIDE	51.56	19.19	48.57	43.71	17.25	38.76
+ RANDOM	51.39	18.89	48.74	43.28	16.87	38.45
<i>II. Low-Rank Methods</i>						
LIN.	50.70	18.48	47.85	43.65	17.12	38.71
<i>III. Learnable Patterns</i>						
LSH	51.95	19.36	48.85	44.74	18.07	39.76
SINKHORN	53.00*	20.05*	50.25*	45.10	18.40*	40.11*
Enc-dec variants w/ full encoder attn.						
LIN.	47.79	14.93	45.15	45.16	17.66	40.25
HEPOS (ours)	51.05*	19.44*	48.51*	45.80*	18.61*	40.69*
Enc-dec variants w/ Sinkhorn encoder attn.						
LIN.	42.90	12.86	40.32	44.84	17.65	39.98
HEPOS (ours)	51.34*	19.09*	48.73*	44.85	18.19*	39.91

Table 3: Results on evaluating encoder and encoder-decoder attentions on input of the same length. Best ROUGE scores of *fixed patterns*, *learnable patterns*, and *enc-dec attentions* are in **red**, **orange**, and **purple**, respectively. *: significantly better than comparison(s) using the same encoder or enc-dec attention (approximation randomization test, $p < 0.0005$).

2020), we combine GLOBAL, STRIDE, and RANDOM with WINDOW and ADASPAN, where we set $g = n^2/s = r = 128$ for a fair comparison. We adapt Linformer to encoder-decoder attentions to compare with HEPOS, where we use $s_h = n/k = 4$ for all experiments. Finally, we report results using FULL, i.e., the original, encoder and encoder-decoder attentions.

Results. Among all *encoder variants*, *learnable patterns* perform the best, approaching the performance of full attentions on both GovReport and PubMed, as shown in Table 3. Within learnable patterns, Sinkhorn attention consistently obtains better ROUGE scores. Moreover, combining techniques in fixed patterns is more effective than simply using window-based sparse attentions, though with an increased memory cost.

For *encoder-decoder attentions*, HEPOS consistently yields higher ROUGE scores than Linformer on both datasets, using either full or Sinkhorn encoder. Notably, coupled with a Sinkhorn attention, our model’s performance matches the variant using

System (MAXLEN)	GovReport			PubMed		
	R-1	R-2	R-L	R-1	R-2	R-L
Baselines						
PEGASUS (1024)	–	–	–	45.97	20.15	41.34
TLM (full)	–	–	–	42.13	16.27	39.21
SEAL (full)	–	–	–	46.50	20.10	42.20
DANCER (full)	–	–	–	46.34	19.97	42.42
BIGBIRD (3072)	–	–	–	46.32	20.65	42.33
Encoder variants w/ full enc-dec attn.						
FULL (1024)	52.83	20.50	50.14	45.36	18.74	40.26
STRIDE (4096)	54.29	20.80	51.35	46.95	19.98	41.67
LIN. (3072)	44.84	13.87	41.94	43.69	16.35	38.66
LSH (4096)	54.75	21.36	51.27	47.54	20.79	42.22
SINKHORN (5120)	55.45	21.45	52.48	47.96	20.78	42.53
Encoder variants w/ HEPOS enc-dec attn. (ours)						
LSH (7168)	55.00	21.13	51.67	48.12	21.06	42.72
SINKHORN (10240)	56.86	22.62	53.82	47.93	20.74	42.58

Table 4: ROUGE scores for models trained on the same GPU. SINKHORN with HEPOS enc-dec attention and LSH with HEPOS both read more text and obtain significantly better scores than other models on GovReport and PubMed ($p < 0.0005$).

System (MAXLEN)	R-1	R-2	R-L
Baselines			
PEGASUS (1024)	44.21	16.95	38.83
TLM (full)	41.62	14.69	38.03
SEAL (full)	44.3	18.0	39.3
DANCER (full)	45.01	17.60	40.56
BIGBIRD (3072)	46.63	19.02	41.77
Encoder variants w/ HEPOS enc-dec attn. (ours)			
LSH (7168)	48.24	20.26	41.78
SINKHORN (10240)	47.87	20.00	41.50

Table 5: Automatic evaluation on arXiv. Our best model yields better ROUGE scores than previous state-of-the-art models.

full encoder attention, implying the effectiveness of HEPOS on both identifying the salient content and capturing the global context.

6.3 Reading More Input Boosts Informativeness

We investigate whether processing more words generates more informative summaries.

Comparisons include recent top-performing *abstractive* models: PEGASUS (Zhang et al., 2019), a large pre-trained summarization model with truncated inputs; TLM (Pilault et al., 2020), DANCER (Gidiotis and Tsoumakas, 2020), and SEAL (Zhao et al., 2020), all of them using hybrid extract-then-abstract methods; and BIGBIRD (Zaheer et al., 2020), which combines sliding window,